

Bike Sharing Demand Prediction using Machine Learning

Project Title

Bike Sharing Demand Prediction using Machine Learning

Problem Statement

The goal of this project is to predict bike rental demand based on factors such as weather, season, temperature, humidity, and working days. Accurate prediction helps bike rental companies manage resources efficiently.

Aim / Objective

To build a machine learning regression model that predicts the number of rented bikes and identifies the factors affecting bike demand.

Dataset Information

The dataset contains historical bike rental records along with environmental and seasonal information such as temperature, humidity, wind speed, holiday, season, and weather conditions.

Features and Target Variable

Input Features: Temperature, Humidity, Windspeed, Season, Weather, Workingday, Holiday, etc.

Target Variable: Bike rental count

Technologies / Libraries Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, XGBoost

Project Workflow / Pipeline

1. Data Collection
2. Data Cleaning
3. Exploratory Data Analysis (EDA)
4. Feature Engineering
5. Data Preprocessing
6. Model Training
7. Model Evaluation
8. Model Comparison and Final Selection

Data Cleaning and Preprocessing

Checked missing values and duplicates, handled data types, performed train-test split, and prepared data for machine learning models.

Exploratory Data Analysis (EDA) Insights

- Temperature positively affects bike rentals.
- High humidity reduces bike demand.
- Working days and seasons strongly influence rental count.
- Some features showed strong correlation with the target variable.
- Data visualization helped identify important trends and outliers.

Machine Learning Models Used

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- XGBoost Regressor

Model Comparison Table

Model	MAE	RMSE	R ² Score	Performance
Linear Regression	104.80	139.21	0.388	Poor Performance
Decision Tree	34.18	58.24	0.893	Good Performance
Random Forest	24.90	42.07	0.944	Very Good Performance
XGBoost	25.00	40.24	0.949	Best Performance

Best Performing Model with Reason

XGBoost Regressor achieved the best performance with an R² score of 0.949 and the lowest RMSE value. It performed better because it handles complex patterns, reduces overfitting, and improves prediction accuracy using boosting techniques.

Evaluation Metrics Explanation

- MAE (Mean Absolute Error): Measures average prediction error.
- RMSE (Root Mean Squared Error): Measures how far predictions are from actual values.
- R² Score: Shows how well the model explains the data variation. Higher values indicate better performance.

Important Visualizations Explanation

- Heatmap: Showed correlation between features.
- Boxplots: Helped identify outliers.
- Histograms: Displayed feature distribution.
- Scatter plots: Showed relationships between variables and bike demand.

Final Result / Conclusion

The project successfully predicted bike rental demand using machine learning models. Among all models, XGBoost achieved the best results with high prediction accuracy and low error values. The project demonstrates how data analysis and machine learning can help businesses improve decision-making.

Challenges Faced

- Understanding relationships between features.
- Interpreting regression evaluation metrics.
- Handling outliers and feature importance.
- Comparing multiple machine learning models.

Future Improvements

- Perform hyperparameter tuning.
- Use larger and real-time datasets.
- Deploy the model as a web application.
- Try advanced deep learning models for better accuracy.

Key Skills Used

Data Cleaning, Data Visualization, EDA, Feature Engineering, Regression Modeling, Model Evaluation, Machine Learning, Python Programming